

Technique 6: D Field and H Field

MP

Teach You Physics Hand In Hand

July 2026

1 Solution to problem in the previous post

Level of difficulty: 4/10

Section A

A.1 [2 points]

RC discharge differential equation:

$$\frac{dQ}{dt} = -\frac{Q}{r_w C} = -\frac{Q}{\tau}$$

Separate variables and integrate:

$$\int_{Q_0}^Q \frac{dQ}{Q} = -\frac{1}{\tau} \int_0^t dt \Rightarrow \ln \frac{Q}{Q_0} = -\frac{t}{\tau}$$

Thus the charge on the upper plate is

$$Q(t) = Q_0 e^{-t/\tau} \quad \text{1 point}$$

The total current in the wire (flowing downward) is

$$I_{\text{total}}(t) = -\frac{dQ}{dt} = \frac{Q_0}{\tau} e^{-t/\tau} \equiv I_0 e^{-t/\tau} \quad \text{1 point}$$

A.2 [4 points] Consider a small ring on the plate. We have:

$$dQ = 2\pi r \sigma dr \quad \text{1 point}$$

$$I = 2\pi \frac{d\sigma}{dt} \int_0^r r dr = \pi \epsilon_0 \frac{dE}{dt} r^2 \quad \text{1 point}$$

$$I_{\text{total}} = \pi \epsilon_0 \frac{dE}{dt} R^2 \quad \text{1 point}$$

Partial credit: [3 points] Consider the charge enclosed within an imaginary cylinder of radius r :

$$Q_{\text{enc}}(r, t) = \sigma(t)\pi r^2 = \frac{Q(t)}{\pi R^2}\pi r^2 = Q(t)\frac{r^2}{R^2} \quad \text{1 point}$$

The radial conduction current flowing toward the axis is the rate of decrease of this enclosed charge:

$$I_{\text{cond}}(r, t) = -\frac{dQ_{\text{enc}}}{dt} = -\frac{d}{dt}\left(Q(t)\frac{r^2}{R^2}\right) \quad \text{1 point}$$

Therefore

$$I_{\text{cond}}(r, t) = I_{\text{total}}(t)\frac{r^2}{R^2} \quad \text{1 point} \quad \text{1 point}$$

A.3 [1 point]

Apply Ampère-Maxwell law to a circular loop of radius r in the gap, we get the displacement current through the loop is

$$\varepsilon_0 \frac{d\Phi_E}{dt} = \varepsilon_0 \frac{d}{dt}(E(t)\pi r^2) = \varepsilon_0 \pi r^2 \frac{d}{dt}\left(\frac{Q(t)}{\varepsilon_0 \pi R^2}\right) = \frac{r^2}{R^2} \frac{dQ}{dt} = -I_{\text{total}}(t)\frac{r^2}{R^2}$$

The conduction current through the loop is the wire current $I_{\text{total}}(t)$, so

$$B_\phi 2\pi r = \mu_0 I_{\text{total}}(t) \left(1 - \frac{r^2}{R^2}\right)$$

Hence

$$B_\phi(r, t) = \frac{\mu_0 I_{\text{total}}(t)}{2\pi r} \left(1 - \frac{r^2}{R^2}\right) \quad (\text{anti-clockwise for } Q_0 > 0) \quad \text{1 point}$$

Section B

B.1 [2 points]

The electric field between the plates (downward, $z > 0$):

$$\vec{E}(t) = \frac{Q(t)}{\varepsilon_0 \pi R^2} \hat{z} = E_0 e^{-t/\tau} \hat{z} \quad \text{1 point}$$

With $\vec{v} = \dot{r}\hat{r} + r\dot{\phi}\hat{\phi} + \dot{z}\hat{z}$ and $\vec{B} = B_\phi\hat{\phi}$, the force components are

$$F_r = e\dot{z}B_\phi, \quad F_\phi = 0, \quad F_z = -eE - e\dot{r}B_\phi \quad \text{1 point}$$

B.2 [3 points] We have:

$$\frac{d}{dt}(mr^2\dot{\phi}) = 0$$

Hence angular momentum about the z -axis is conserved. **1 point** Therefore the motion is purely in the (r, z) plane. Using the force components:

$$m\ddot{r} = F_r = e\dot{z}B_\phi \Rightarrow \ddot{r} = \frac{e}{m}\dot{z}B_\phi \quad \text{1 point}$$

and

$$m\ddot{z} = F_z = -eE - e\dot{r}B_\phi \Rightarrow \ddot{z} = -\frac{e}{m}E - \frac{e}{m}\dot{r}B_\phi \quad \text{1 point}$$

Section C

C.1 [4 points]

Let $u = \dot{r}$ and $w = \dot{z}$. Under the constant-field approximation, the coupled equations are

$$\dot{u} = \frac{eB_0}{m}w, \quad \dot{w} = -\frac{eE_0}{m} - \frac{eB_0}{m}u$$

Differentiate the first equation and substitute the second:

$$\ddot{u} + \left(\frac{eB_0}{m}\right)^2 u = -\frac{eE_0}{m} \frac{eB_0}{m}$$

Solving with $u(0) = 0$, $w(0) = 0$ gives

$$u(t) = \frac{E_0}{B_0} \left(\cos \frac{eB_0}{m}t - 1 \right) \quad \text{2 points}, \quad w(t) = -\frac{E_0}{B_0} \sin \frac{eB_0}{m}t \quad \text{2 points}$$

C.2 [4 points] Integrating:

$$r(t) = r_0 + \frac{mE_0}{eB_0^2} \left(\sin \frac{eB_0}{m}t - \frac{eB_0}{m}t \right) \quad \text{1 point}$$

$$z(t) = \frac{mE_0}{eB_0^2} \left(\cos \frac{eB_0}{m}t - 1 \right) \quad \text{1 point}$$

Decompose the displacement:

$$r(t) = r_0 + \underbrace{\frac{mE_0}{eB_0^2} \sin \frac{eB_0}{m}t}_{\text{circular}} - \underbrace{\frac{E_0}{B_0}t}_{\text{drift}}, \quad z(t) = -\underbrace{\frac{mE_0}{eB_0^2} \left(1 - \cos \frac{eB_0}{m}t \right)}_{\text{circular}}$$

The cyclotron radius is

$$R_c = \frac{mE_0}{eB_0^2}$$

with centre at $(0, -R_c)$ in the (r, z) plane. 1 point

The drift velocity is

$$\vec{v}_{\text{drift}} = -\frac{E_0}{B_0} \hat{r}$$

Verify with the classical formula:

$$\frac{\vec{E} \times \vec{B}}{B^2} = \frac{E_0 \hat{z} \times B_0 \hat{\phi}}{B_0^2} = -\frac{E_0}{B_0} \hat{r}$$

Thus the drift velocity matches exactly. 1 point

2 Basic relations between E,B,D,H,P,M

Net Effect	Input	Medium Response
$\epsilon_0 E$	D	$-P$
$\frac{1}{\mu_0} B$	H	$\pm M$

All you have to know is:

$$\text{Net Effect} = \text{Input} + \text{Medium Response}$$

which is this set of equations:

$$\epsilon_0 E = D - P \quad (1)$$

$$\frac{1}{\mu_0} B = H \pm M \quad (2)$$

From equation 1, the electric field input is given out by density of **free charges**, which is the **divergence** taken on **D field** and the original cause of D field:

$$\nabla \cdot D = \rho_{\text{free}}$$

The medium tends to oppose the input field by setting up another electric field with **bound charges**, whose density is the **divergence** taken on **P** and the original cause of polarization:

$$-\nabla \cdot P = \rho_{\text{bound}}$$

Obviously, the **net electric field** or **resultant electric field** is given by:

$$E = \frac{1}{\epsilon_0} (D - P)$$

And the **effective charge density** is given by:

$$\nabla \cdot (\epsilon_0 E) = \rho_{\text{net}} = \rho_{\text{free}} - \rho_{\text{bound}} = \nabla \cdot D - \nabla \cdot P$$

At interfaces between different media, the normal components satisfy the boundary conditions:

$$\sigma_{\text{free}} = \hat{n} \cdot (D_1 - D_2), \quad \sigma_{\text{bound}} = -\hat{n} \cdot (P_1 - P_2), \quad \sigma_{\text{net}} = \hat{n} \cdot (\epsilon_0 E_1 - \epsilon_0 E_2)$$

where $\hat{n}$ is the unit normal pointing from medium 1 to medium 2. These are simply the surface analogues of the divergence relations above, with $\sigma_{\text{net}} = \sigma_{\text{free}} - \sigma_{\text{bound}}$.

From equation 2, the magnetic field input is given out by density of **free current**, which is the **curl** taken on **H field** and the original cause of H field:

$$\nabla \times H = J_{\text{free}}$$

The medium tends to strengthen or oppose the input field by setting up another current of **medium current**, whose density is the **curl** taken on **M** and the original cause of magnetization:

$$\nabla \times M = J_{\text{bound}}$$

Obviously, the **net magnetic field** or **resultant magnetic field** is given by:

$$B = \mu_0(H \pm M)$$

And the **effective current density** is given by:

$$\nabla \times \left(\frac{1}{\mu_0} B \right) = J_{\text{net}} = J_{\text{free}} - J_{\text{bound}} = \nabla \times H \pm \nabla \times M$$

The sign M depends on the property of the medium. For the magnetic fields, the normal component of B is continuous, while the tangential components of H and M jump by surface currents:

$$\hat{n} \cdot (B_1 - B_2) = 0, \quad \hat{n} \times (H_1 - H_2) = K_{\text{free}}, \quad \hat{n} \times (M_1 - M_2) = K_{\text{bound}}$$

where K_{free} and K_{bound} are the free and bound surface current densities, respectively. The net surface current is then

$$\hat{n} \times \left(\frac{B_1}{\mu_0} - \frac{B_2}{\mu_0} \right) = K_{\text{net}} = K_{\text{free}} - K_{\text{bound}}.$$

3 Problem

Transmission Electron Microscopy (TEM) no optics :)

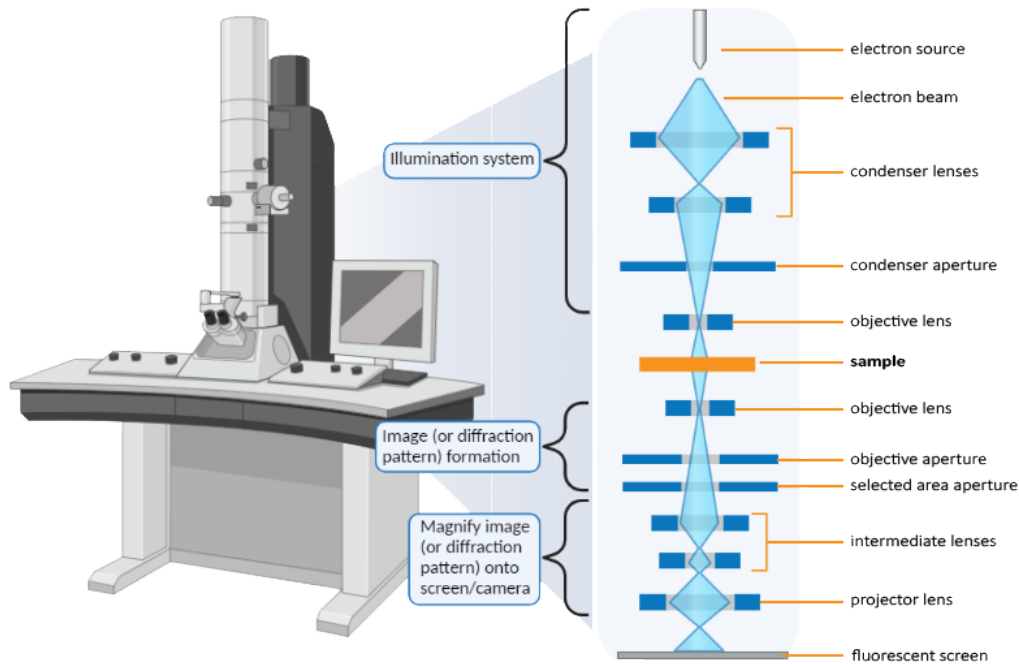
[20 points] A curved solenoid is put into a random magnetic field $B(t)$ with a random orientation. The radius of a coil is r , the coil density is n , the resistance is R . Two ends (a and b) of the solenoid is attached to an unknown circuit in a black box and a complete circuit is formed. Given that there is no power supply in the box. Assume the coils are not magnetized. You can assume the curvature of solenoid and thickness of coils are small such that $\hat{n}dl = d\vec{l}$, where $\hat{n}$ is area vector of a coil.

Part A: Induced EMF in TEM

In **Part A**, you only have to treat the black box as a conducting wire.

A1 [3 points] Now, the magnetic field starts to increase from 0 and the solenoid remains at original position. Find the induced EMF in the solenoid. Leave your answer as integral involving vector integral if necessary.

A2 [2 points] Find the total amount of current flowed through the solenoid. Hence, write down the expression of charge flowed through. Leave your answer as integral involving vector integral if necessary.



The picture is just for fun! In modern transmission electron microscopy (TEM), spherical aberration is one of the primary factors limiting spatial resolution. The hexapole aberration corrector has become the industry standard for correcting this aberration, enabling sub-angstrom imaging in cutting-edge instruments manufactured by companies such as Thermo Fisher Scientific and JEOL.

At the heart of the hexapole corrector lies a set of specially shaped electromagnetic coils. Unlike simple circular solenoids, these coils must follow non-circular curved paths to generate the required hexapole magnetic field distribution. The planar involute geometry used in this problem, with its smoothly varying curvature and open path, where $\theta_b - \theta_a \neq 2\pi k$, is a realistic approximation of such coil designs. The magnetic field configuration

$$\vec{H}(\vec{r}, t) = H_0(t) \begin{pmatrix} \cos 2\theta(\vec{r}) \\ \sin 2\theta(\vec{r}) \\ 0 \end{pmatrix},$$

directly corresponds to the hexapole symmetry required for aberration correction.

A3 [7 points] Modify the expression of current in A.2 under the above conditions. Given that The solenoid is wound as:

$$\vec{r}(\theta) = R_0 \theta \begin{pmatrix} \cos \theta \\ \sin \theta \\ 0 \end{pmatrix}$$

You may use the substitution:

$$A = [\sin \theta - \theta \cos \theta]_{\theta_a}^{\theta_b} = (\sin \theta_b - \theta_b \cos \theta_b) - (\sin \theta_a - \theta_a \cos \theta_a)$$

Part B: Circuit in TEM

In a modern transmission electron microscope, the hexapole aberration corrector employs a curved solenoid to generate a time-varying magnetic field. This solenoid can be modeled as an equivalent inductance L in series with a resistance R . The solenoid is connected to a black box that contains a capacitor C and a negative differential resistance device, such as a tunnel diode, whose differential resistance is $-R_0$ with $R_0 > 0$, provided the instantaneous current through the device remains within its specified operating range. To maintain the device in its negative resistance region, an ideal DC current source I_0 is connected to it through an ideal RF choke. The RF choke presents an open circuit to AC signals while offering zero resistance to DC. Consequently, the AC dynamics of the loop, which consists of the solenoid with inductance L and resistance R , the capacitor C , and the negative differential resistance $-R_0$, are completely unaffected by the DC bias network.

B1 [5 points] Write the differential equation governing the AC current $i(t)$ in the loop. Given the initial conditions $i(0) = 0$ and $\dot{i}(0) = \beta \neq 0$, solve for $i(t)$ for $t > 0$ in the case where

$$\left(\frac{R - R_0}{2L} \right)^2 < \frac{1}{LC}.$$

Express your answer in terms of L , R , R_0 , C , and β .

B2 [3 points] Find the condition on R_0 for which the circuit exhibits undamped oscillations. What is the oscillation frequency f_0 in this case? If the NDR device can only maintain its negative resistance when the AC voltage amplitude across it does not exceed $V_{\max}$, what is the maximum average power $\bar{P}_{\max}$ that can be delivered to the circuit by the NDR during sustained oscillations? Express your answer in terms of $V_{\max}$ and R_0 .

The solution will be uploaded in the next post and MP love you all :)